Naïve Bayes 2

Saturday, March 9, 2024 9:04 PM

Text → NB < 0 / 1

Text

Cleaning : ① Token
② Lower
③ Punctuation
④ Root (eat)
⑤ Stopwords

$P(y=1 \mid text)$
$P(y=0 \mid text)$ } compare

$$P(y=1 \mid text) = \frac{P(text \mid y=1) \times P(y=1)}{\cancel{P(text)}}$$ common

$$P(w_1 w_2 w_3 \ldots w_d \mid y=1) \times P(y=1)$$

words are Independent based on cond?

$$P(w_1 \mid y=1) \times P(w_2 \mid y=1) \times \ldots P(w_d \mid y=1) \times P(y=1)$$

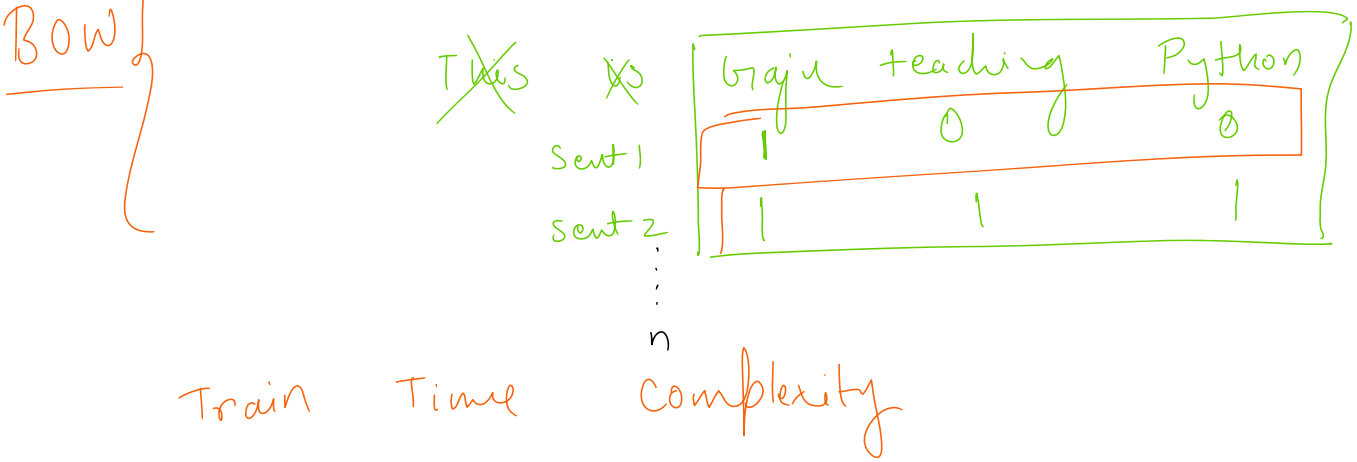
2 Sentence

2 Sentence

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Gaju teaching Python

BOW




Train Time complexity

$P(y=1)$

$P(y=0)$

$w_1 \ldots\ldots w_{100}$

$(y=1)(y=0)$

$P(w_i \mid y=1) \ldots d$

$P(w_i \mid y=0) \ldots d$

n of sentence

$O(2nd + 2)$

$O(nd)$

Test

$w_1 \; w_2 \; w_3 \ldots\ldots w_K$

$y=1 \quad y=0$

$w_1$ $w_2$ $w_3$ . . . . $w_k$

$y=1$ $y=0$

$O(2k)$

$O(k)$

---

$P(y=0)$

$P(y=1)$

like

$w_1$ . . . - $w_d$

$O(2d)$

↓

$O(d)$

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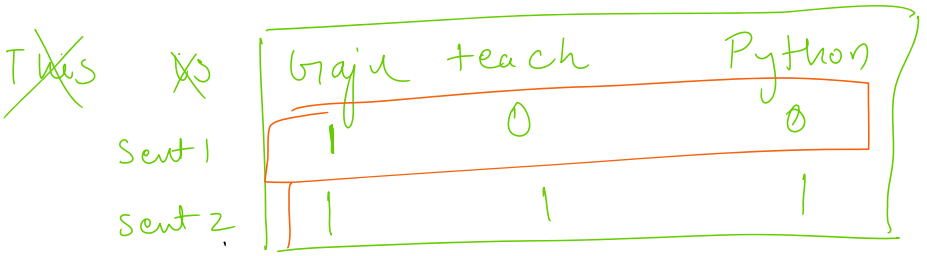
# Laplace Smoothing

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} Training

BOW }



Syed teach Python

Python → $\frac{\text{no of Times Python appred in } y=1}{\text{no of sentences in class 1}}$

$$P(y=1) \times P(w_1 \mid y=1) \times P(w_2 \mid y=1) \cdots P(w_d \mid y=1)$$

$$P(y=1) \times \frac{P(w_1 \mid y=1) \times P(w_2 \mid y=1) \cdots}{P(w_d \mid y)} = \frac{0}{2} \approx 0$$

not in traing

$$P(y=0) \times \quad = 0$$

N P M × ###

Laplace Smoothing / Additive Smoothing

$$P(w_d \mid y=1) = \frac{n_d + \alpha}{n + \alpha \cdot C}$$

C → distinct value

↳ distinct value class can take

1 / 0 } 2

100 text msg → y=1 - spam.

$\alpha = 1 \qquad C = 2$

$$P(w' \mid y=1) = \frac{0+1}{100+1\times 2} = \frac{1}{102} \quad 0.0098$$

## Hyperparameter Tuning

$\alpha$ is very large

$$\frac{n_j + \alpha}{n + \alpha \cdot C}$$

$y =$ 1 spam — (100) msg

0 non spam — 98 msg

$w_1$ - study → (10)

$w_1$ - study → 7

$$P(w_1 \mid y=1) = \frac{10}{100} = (0.1)$$ } without laplace

$\alpha = 1$
$C = 2$

$= \frac{10+1}{100+1\times 2} = 0.1$

α descent

$\alpha = 10000 \quad = \frac{10 + 10000}{100 + 10000 \times 2} = \frac{1}{2} \leftarrow$

$P(\omega \mid y=0) \quad \frac{7 + 10000}{98 \quad 20000} = \frac{1}{2} \leftarrow$

$\rightarrow P(y=1) \times \text{likelihood} \times 0.7$ ✓
$\rightarrow P(y=0) \times \text{likelihood} \times 0.3$
} Imbalanced

$\alpha \uparrow \rightarrow$ Majority class $\rightarrow$ Underfit

$\alpha \downarrow \rightarrow \frac{1 + 0.0001}{100 + 0.0001 \times 2} \longrightarrow \frac{1}{100} \rightarrow$ overfit

Hence, to summarize, we have a **hyper parameter (α)** using which we can **control the extent of overfitting or underfitting.**

- As **alpha increases**, we begin **underfitting** the data
- As **alpha decreases**, we

begin **overfitting** the data

# Impact of Imbalance Dataset

$P(y=1) = 0.7$ — 70/100

$P(y=0) = 0.3$ — 30/100

Reblance — upsampling

$P(y=1|text) = P(y=1) \times \Pi$ likelihood.

0.07

0.1

$P(y=0|text) = P(y=0) \times \Pi$ likelihood

0.2

0.06

50/100

50/100

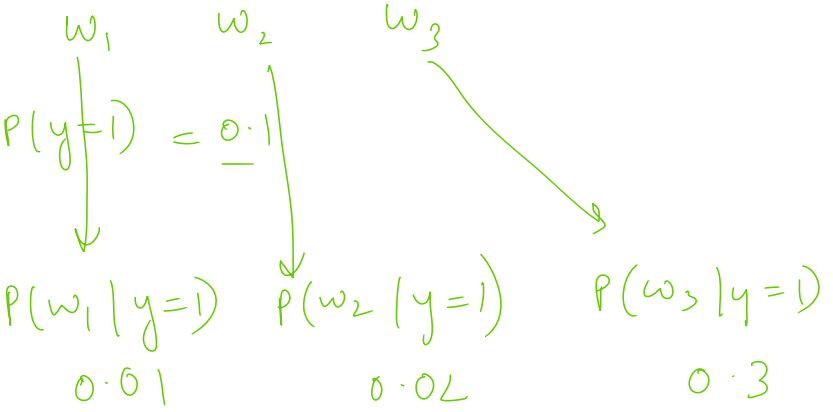
- Hyperparameter tuning — $\alpha$
- Impact of Imbalance
- underflow

- Features Importance

$P(y=1 \mid text)$ — d unique words

→ $P(y=1)$ Prior 0-1

→ $\prod_{j=1}^{d} P(w_j \mid y=1)$ 0-1 } ∞




$= 0.000123$

d ↑ increases → decreasing → underflow

Float64

16 → $w_1$ $w_2$ $w_3$ } Loss of Information

## ② Log — Scaling

$$P(y=1 \mid w_1 w_2 \dots w_d) = P(y=1) \times \prod_{j=1}^{d} P(w_j \mid y=1)$$

$$\log\left(P(y=1 \mid w_1 w_2 \dots w_d)\right) = \log\left(P(y=1) \times \prod_{j=1}^{d} P(w_j \mid y=1)\right)$$

$$\log ab = \log a + \log b$$

$$= \log\left(P(y=1)\right) + \sum \log\left(P(w_j \mid y=1)\right)$$

0.1 + 0.01 + 0.02

$$\log\left(P(y=1)\right) + \log\left(P(w_1 \mid y=1)\right) + \log P(w_2 \mid y=1)$$

why Naive bayes is Naive ?

$w_1$ $w_2$ are ~~independent~~

The Naivity of Naive Bayes is assuming that the words ⬛⬛*wi* and ⬛⬛*wj* are independent of each other only on the condition when y=1 or y=0.

# Features Importance

$w_1$ $w_2$ $w_3$

which word is important

you won a choc → $P(y=1) > P(y=0)$

you won a choc → Spam

$P(y=1)$ $P(y=1)$ $P(y=1)$ $P(y=1)$ → Prior

$P(\text{you} \mid y=1)$ $P(\text{won} \mid y=1)$ $P(\text{a} \mid y=1)$ $P(\text{choc} \mid y=1)$ → Likelihood

1 ↓ 0.8 | ↓ 0.7

0.01 0.01

Most imp feature

Impact outlier

Sycl — ~~Training~~
↳ Test

$w_j$ = only once 10000 training

$= n_j = 1$
$n_1 = 10000$ } rare

large $\alpha$

{ $\frac{1}{1000}$ 0.0001

$\frac{1 * 100}{1000 + 100 \times 2} = \frac{100}{} = 0.0$ }

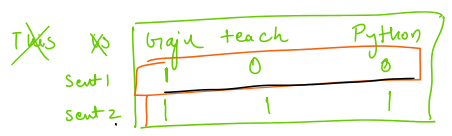
Threshold < 10

$\frac{7}{1000} - \frac{0}{1000}$

## Bernoulli

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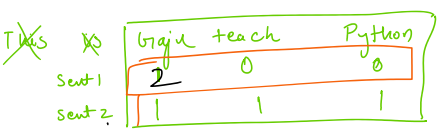
OW {

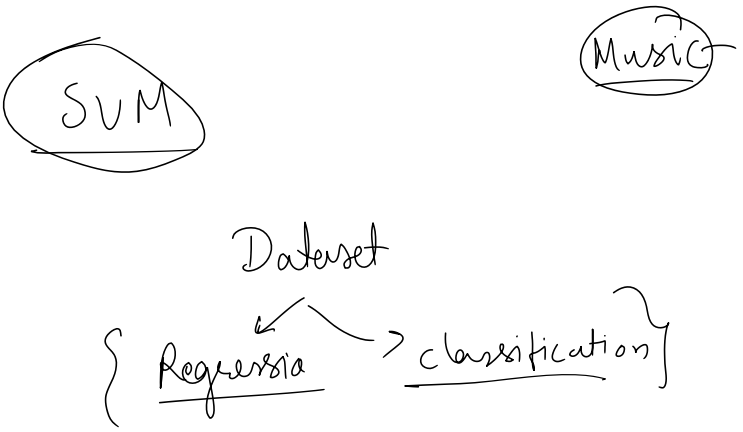



## Multinomial

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SVM
Music
Dataset
Regressia
classification